

MA4110/MA5020: Computational Methods for Fluid Flow

Tutorial Sheet 2

Finite Difference Schemes, Truncation Error, Stability, and Modified Equations

Instructor: Biswarup Biswas

Question 1: Order of Accuracy of Finite Difference Formulas

- (a) Using Taylor series expansions, find the leading truncation error and order of accuracy of:

$$u'(x_j) \approx \frac{-3u_j + 4u_{j+1} - u_{j+2}}{2\Delta x}$$

- (b) Determine constants a, b, c such that $u''(x_j) \approx \frac{a u_{j-1} + b u_j + c u_{j+1}}{\Delta x^2}$ is second-order accurate.

Question 2: Local Truncation Error of the Upwind Scheme

Consider the First-Order Upwind (FOU) scheme for $u_t + a u_x = 0$ ($a > 0$): $\frac{u_j^{n+1} - u_j^n}{\Delta t} + a \frac{u_j^n - u_{j-1}^n}{\Delta x} = 0$.

- (a) Expand u_j^{n+1} and u_{j-1}^n in Taylor series about (x_j, t^n) to determine the local truncation error τ_j^n .
- (b) Express τ_j^n purely in spatial derivatives using $u_t = -a u_x$ and state the formal order of accuracy.

Question 3: Hand Calculation: 1D Advection Equation

Consider the advection problem $\frac{\partial u}{\partial t} + 2 \frac{\partial u}{\partial x} = 0$ on $x \in [0, 1]$ with $u(x, 0) = \sin(\pi x)$ and $u(0, t) = 0$. Discretize with $\Delta x = 0.2$ (nodes $x_0 = 0.0, x_1 = 0.2, \dots, x_5 = 1.0$) and time step $\Delta t = 0.05$.

- (a) Calculate the Courant number $\nu = \frac{a\Delta t}{\Delta x}$. Is the First-Order Upwind scheme stable for this choice?
- (b) Using the upwind scheme $u_j^{n+1} = u_j^n - \nu(u_j^n - u_{j-1}^n)$, compute the numerical solution at:
- $t = 0.05$ ($n = 1$) for all nodes $j = 0, 1, 2, 3, 4, 5$.
 - $t = 0.10$ ($n = 2$) for all nodes $j = 0, 1, 2, 3, 4, 5$.
- (c) Compare your computed numerical value u_2^2 at $(x, t) = (0.4, 0.1)$ with the exact solution $u_{\text{exact}}(x, t) = \sin(\pi(x - 2t))$.

Question 4: Hand Calculation: 1D Heat Equation (FTCS)

Solve $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$ on $x \in [0, 1]$ using explicit FTCS: $u_j^{n+1} = u_j^n + r(u_{j+1}^n - 2u_j^n + u_{j-1}^n)$ with $r = \frac{\Delta t}{\Delta x^2}$, boundary conditions $u(0, t) = u(1, t) = 0$, and initial condition $u(x, 0) = 100 \sin(\pi x)$. Use $\Delta x = 0.2$ and $\Delta t = 0.01$.

- (a) Compute the diffusion number r . Is this computation numerically stable?
- (b) Compute the values at internal nodes ($x = 0.2, 0.4, 0.6, 0.8$) at $t = 0.01$ (step 1) and $t = 0.02$ (step 2).
- (c) What would happen to the solution if we chose $\Delta t = 0.03$? Explain why based on stability theory.

Question 5: Von Neumann Stability for Advection Schemes

Using Von Neumann analysis ($u_j^n = G^n e^{ij\xi}$), find the amplification factor $G(\xi)$ and determine the stability condition (range of CFL number $\nu = \frac{a\Delta t}{\Delta x}$) for each scheme applied to $u_t + au_x = 0$ ($a > 0$):

- (a) **FTCS Scheme:** $\frac{u_j^{n+1} - u_j^n}{\Delta t} + a \frac{u_{j+1}^n - u_{j-1}^n}{2\Delta x} = 0$. Show that $|G(\xi)|^2 > 1$ for $\xi \neq 0$ (unconditionally unstable).
- (b) **First-Order Upwind Scheme:** $\frac{u_j^{n+1} - u_j^n}{\Delta t} + a \frac{u_j^n - u_{j-1}^n}{\Delta x} = 0$. Show that $|G(\xi)| \leq 1$ requires $0 \leq \nu \leq 1$.
- (c) **Lax-Friedrichs Scheme:** $u_j^{n+1} = \frac{1}{2}(u_{j+1}^n + u_{j-1}^n) - \frac{\nu}{2}(u_{j+1}^n - u_{j-1}^n)$. Show that it is stable for $|\nu| \leq 1$.

Question 6: Stability of Backward-Time Central-Space (BTCS)

Consider the fully implicit BTCS scheme for $u_t = \alpha u_{xx}$: $\frac{u_j^{n+1} - u_j^n}{\Delta t} = \alpha \frac{u_{j+1}^{n+1} - 2u_j^{n+1} + u_{j-1}^{n+1}}{\Delta x^2}$.

- (a) Derive the amplification factor $G(\xi)$ in terms of $r = \frac{\alpha\Delta t}{\Delta x^2}$ and $\sin^2(\xi/2)$.
- (b) Prove that $|G(\xi)| \leq 1$ for all $r > 0$. What does this imply about the time step Δt ?

Question 7: Modified Equation Analysis of the Upwind Scheme

For First-Order Upwind applied to $u_t + au_x = 0$ ($a > 0$):

- (a) Derive the modified equation up to the second derivative: $\frac{\partial u}{\partial t} + a \frac{\partial u}{\partial x} = \alpha_{\text{num}} \frac{\partial^2 u}{\partial x^2}$, and write α_{num} in terms of $a, \Delta x, \nu$.
- (b) What happens to the artificial viscosity α_{num} when $\nu = 1$? Explain the physical meaning.

Question 8: Dispersion Errors: Lax-Wendroff vs. Beam-Warming

The modified equations for Lax-Wendroff and Beam-Warming schemes are:

$$\text{Lax-Wendroff: } \frac{\partial u}{\partial t} + a \frac{\partial u}{\partial x} = -\frac{a\Delta x^2}{6}(1-\nu^2) \frac{\partial^3 u}{\partial x^3}, \quad \text{Beam-Warming: } \frac{\partial u}{\partial t} + a \frac{\partial u}{\partial x} = +\frac{a\Delta x^2}{6}(2-3\nu+\nu^2) \frac{\partial^3 u}{\partial x^3}$$

- (a) For $\nu = 0.5$, state the sign of the coefficient of u_{xxx} for each scheme.
- (b) Using $c(k) = a - Ck^2$ for $u_t + au_x = -Cu_{xxx}$, determine which scheme makes high-frequency waves travel slower than a and which makes them travel faster than a .
- (c) State which scheme produces **trailing oscillations** (behind the shock) and which produces **leading oscillations** (ahead of the shock).

Question 9: Godunov's Barrier Theorem and Monotonicity

- (a) State the condition on the coefficients c_k of a general linear scheme $u_j^{n+1} = \sum_k c_k u_{j+k}^n$ to guarantee that the scheme is monotonicity-preserving.
- (b) For Lax-Wendroff, the coefficient of u_{j+1}^n is $c_1 = -\frac{\nu(1-\nu)}{2}$. Show that $c_1 < 0$ for all $0 < \nu < 1$. What does this negative coefficient cause near sharp gradients?
- (c) State **Godunov's Barrier Theorem (1959)** in one or two clear sentences.